Java Programming: Program-First Learning Roadmap

A complete 307-program progression from absolute beginner Java through OOP, collections, exceptions, multithreading, JDBC and practical projects.

How to use this roadmap

For every program, you need to understand every keyword, class, object, method, operator, statement and line before moving to the next program.

For every program, you must be able to explain:

- What every keyword, class, method and library/import statement means.
- What every variable, data type, object and reference represents.
- Why a particular primitive type, class or collection is being used.
- What every operator and important symbol does.
- How braces, semicolons and Java blocks work.
- How the program executes line by line.
- What happens when input or data changes.
- What error occurs when a line is deliberately changed or removed.
- For OOP: explain class, object, constructor, inheritance, overriding, polymorphism, abstraction, encapsulation and interfaces.
- For inheritance/interfaces: understand why Java does not allow multiple class inheritance and how interface default-method conflicts are resolved.

Level 1 — Absolute Basics

1. Print Hello World
2. Print your name, age and college name. Print multiple lines using

System.out.println()

4. Print a sentence using System.out.print() and System.out.println()
5. Take a name as input using Scanner and print it
6. Take an integer as input and print it
7. Take two integers and print their sum
8. Perform addition, subtraction, multiplication and division
9. Calculate the square of a number
10. Calculate the cube of a number
11. Calculate the area of a circle
12. Calculate the area and perimeter of a rectangle
13. Convert Celsius to Fahrenheit

14. Swap two numbers using a third variable
15. Swap two numbers without using a third variable

Level 2 — Variables, Data Types & Operators

16. Demonstrate byte, short, int, long, float, double, char and boolean
17. Display the size/range-related information of primitive types where applicable
18. Demonstrate implicit type conversion
19. Demonstrate explicit type casting
20. Demonstrate arithmetic operators
21. Demonstrate relational operators
22. Demonstrate logical operators
23. Demonstrate assignment operators
24. Demonstrate increment and decrement operators
25. Demonstrate the ternary operator
26. Calculate simple interest
27. Calculate compound interest
28. Calculate the average of three numbers
29. Calculate percentage of five subjects
30. Convert seconds into hours, minutes and seconds
31. Calculate total and average marks of a student

Level 3 — Input and Conditional Statements

32. Check whether a number is positive, negative or zero
33. Check whether a number is even or odd
34. Check voting eligibility
35. Find the greater of two numbers
36. Find the greatest of three numbers
37. Check whether a year is a leap year
38. Check whether a character is a vowel or consonant
39. Check whether a character is uppercase or lowercase
40. Check whether a number is divisible by 5
41. Check whether a number is divisible by both 5 and 11
42. Check whether a student has passed or failed
43. Calculate grade based on marks
44. Check whether three sides can form a triangle
45. Determine whether a triangle is equilateral, isosceles or scalene
46. Check whether a number lies within a given range

Level 4 — switch and Menu Programs

47. Create a simple calculator using switch
48. Display the day of the week using a number
49. Display the month name using a number
50. Create a menu-driven area calculator
51. Create a menu-driven unit converter
52. Create a menu-driven student result program
53. Create a menu-driven arithmetic program

Level 5 — Loops

54. Print numbers from 1 to 10
55. Print numbers from 10 to 1
56. Print numbers from 1 to N
57. Print all even numbers from 1 to N
58. Print all odd numbers from 1 to N
59. Find the sum of numbers from 1 to N
60. Find the sum of even numbers from 1 to N
61. Find the sum of odd numbers from 1 to N
62. Print the multiplication table of a number
63. Print multiplication tables from 1 to 10
64. Find the factorial of a number
65. Count the digits of a number
66. Find the sum of digits of a number
67. Extract digits from right to left
68. Reverse a number
69. Check whether a number is a palindrome
70. Find the first and last digit
71. Find the largest digit
72. Find the smallest digit
73. Count occurrences of a particular digit
74. Check whether a number is an Armstrong number
75. Check whether a number is a perfect number
76. Check whether a number is a prime number
77. Print all prime numbers between two numbers
78. Find HCF/GCD of two numbers
79. Find LCM of two numbers
80. Generate the Fibonacci series
81. Check whether a number belongs to the Fibonacci series

Level 6 — Nested Loops & Patterns

82. Print a square of *
83. Print a rectangle of *
84. Print a right-angled triangle of *
85. Print an inverted right-angled triangle
86. Print a number triangle
87. Print an inverted number triangle
88. Print the pattern 1 22 333 4444
89. Print the pattern 1 12 123 1234
90. Print the pattern 1 11 121 1221
91. Print a pyramid of *
92. Print an inverted pyramid
93. Print a number pyramid
94. Print Floyd's triangle
95. Print Pascal's triangle
96. Print a diamond pattern
97. Print a hollow square
98. Print a hollow triangle
99. Print an X pattern using *

Level 7 — Methods / Functions

100. Create a method to add two numbers
101. Create a method to find the square of a number
102. Create a method to check even/odd
103. Create a method to find the greatest of three numbers
104. Create a method to calculate factorial
105. Create a method to check prime number
106. Create a method to reverse a number
107. Create a method to check palindrome
108. Create a method to calculate GCD
109. Create a method to calculate LCM
110. Create a menu-driven calculator using methods
111. Create a student-result program using separate methods
112. Demonstrate method parameters and arguments
113. Demonstrate method overloading
114. Demonstrate methods with return values
115. Demonstrate methods with no return value

Level 8 — Arrays

116. Input and display elements of an array

117. Find the sum of array elements
118. Find the average of array elements
119. Find the largest element in an array
120. Find the smallest element in an array
121. Count even and odd elements
122. Search for an element in an array
123. Count occurrences of an element
124. Reverse an array
125. Copy one array into another
126. Merge two arrays
127. Find the second-largest element
128. Sort an array in ascending order
129. Sort an array in descending order
130. Remove duplicate elements
131. Insert an element at a specified position
132. Delete an element from a specified position
133. Find common elements between two arrays
134. Perform matrix addition
135. Perform matrix subtraction
136. Perform matrix multiplication
137. Find the transpose of a matrix

Level 9 — Strings

138. Create and display a String
139. Find the length of a String
140. Reverse a String
141. Check whether a String is a palindrome
142. Count vowels and consonants
143. Count digits, spaces and special characters
144. Convert lowercase characters to uppercase
145. Convert uppercase characters to lowercase
146. Count the frequency of each character
147. Remove spaces from a String
148. Find the largest word in a sentence
149. Count the number of words in a sentence
150. Find a word inside a sentence
151. Replace one word with another
152. Check whether two Strings are anagrams
153. Compare two Strings using equals()

154. Compare Strings ignoring case
155. Demonstrate StringBuilder
156. Reverse a String using StringBuilder

Level 10 — Object-Oriented Programming Fundamentals

157. Create a Student class and create its objects
158. Create a class with fields and methods
159. Use constructors to initialize objects
160. Create a parameterized constructor
161. Demonstrate constructor overloading
162. Demonstrate this keyword
163. Create getters and setters
164. Demonstrate encapsulation
165. Create a BankAccount class
166. Create an Employee class
167. Create a Book class
168. Create an array of objects
169. Search an object by an ID
170. Sort an array of objects by a field
171. Create a class-based calculator

Level 11 — Inheritance, Polymorphism & Interfaces

172. Create a simple single-inheritance program
173. Demonstrate multilevel inheritance
174. Demonstrate hierarchical inheritance
175. Demonstrate method overriding
176. Demonstrate runtime polymorphism using a parent reference
177. Demonstrate the super keyword
178. Demonstrate constructor behavior in inheritance
179. Demonstrate final methods and final classes
180. Create an abstract class with an abstract method
181. Create a concrete subclass of an abstract class
182. Create an interface and implement it
183. Implement multiple interfaces in one class
184. Demonstrate interface default methods
185. Demonstrate interface static methods
186. Create two interfaces with the same default method and resolve the conflict explicitly
187. Create a class implementing two interfaces that declare the same method
188. Model a diamond-shaped inheritance structure using interfaces and resolve the default-method ambiguity

189. Create a Vehicle interface and implement it in multiple classes
190. Demonstrate polymorphism through an interface reference
191. Compare abstract classes and interfaces through a working program

Level 12 — Packages, Access Modifiers & Exceptions

192. Create and use a package
193. Import a class from another package
194. Demonstrate public access
195. Demonstrate private access
196. Demonstrate protected access
197. Demonstrate package-private access
198. Create a program showing access across packages
199. Demonstrate try and catch
200. Handle division by zero
201. Handle invalid input using exceptions
202. Use multiple catch blocks
203. Use finally
204. Use throw
205. Use throws
206. Create a custom exception
207. Create a program with nested try blocks
208. Create a user-defined exception for invalid marks

Level 13 — Collections Framework

209. Create and traverse an ArrayList
210. Add, remove and update ArrayList elements
211. Search an ArrayList
212. Sort an ArrayList
213. Reverse an ArrayList
214. Remove duplicates from an ArrayList
215. Create and traverse a LinkedList
216. Create and use a HashSet
217. Find union/intersection-like results using sets
218. Create and use a TreeSet
219. Create and use a HashMap
220. Add, update and remove HashMap entries
221. Search a HashMap by key
222. Count word frequencies using HashMap
223. Sort a map by keys using TreeMap

224. Use Iterator
225. Use ListIterator
226. Compare ArrayList and LinkedList with simple programs
227. Compare HashSet and TreeSet with simple programs

Level 14 — Generics, Wrapper Classes & Enums

228. Demonstrate wrapper classes
229. Convert between primitive values and wrapper objects
230. Demonstrate autoboxing and unboxing
231. Create a generic class
232. Create a generic method
233. Use a generic ArrayList
234. Create and use an enum
235. Use enum values in a switch statement
236. Create an enum with fields and methods

Level 15 — File Handling & Serialization

237. Create and write to a text file
238. Read and display a text file
239. Append data to a text file
240. Count characters in a file
241. Count words in a file
242. Count lines in a file
243. Copy one file into another
244. Search for a word in a file
245. Store student records in a file
246. Read student records from a file
247. Update a record in a file
248. Demonstrate BufferedReader
249. Demonstrate BufferedWriter
250. Demonstrate FileInputStream and FileOutputStream
251. Serialize an object
252. Deserialize an object

Level 16 — Multithreading

253. Create a thread by extending Thread
254. Create a thread by implementing Runnable
255. Start multiple threads
256. Demonstrate thread sleep

257. Demonstrate thread join
258. Demonstrate thread priority
259. Create a synchronized method
260. Create a synchronized block
261. Demonstrate a race condition with a simple counter
262. Create a producer-consumer style example using basic synchronization

Level 17 — Java 8+ Functional Features

263. Create a lambda expression
264. Use lambda expressions with a functional interface
265. Use Predicate
266. Use Consumer
267. Use Supplier
268. Use Function
269. Use method references
270. Create a stream from a collection
271. Filter a collection using streams
272. Map elements using streams
273. Sort a collection using streams
274. Find maximum and minimum using streams
275. Count elements using streams
276. Group data using Collectors.groupingBy
277. Calculate summary statistics using streams
278. Use Optional for a possibly missing value

Level 18 — JDBC / Database Basics

279. Connect a Java program to a database using JDBC
280. Create a database table through JDBC
281. Insert a record using JDBC
282. Read records using JDBC
283. Update a record using JDBC
284. Delete a record using JDBC
285. Use PreparedStatement
286. Search records using PreparedStatement
287. Create a simple student database application using JDBC

Level 19 — Small Projects

288. Number guessing game
289. Rock-paper-scissors game

290. Simple quiz program
291. Student marks management system
292. Student record management system
293. Employee record management system
294. Simple banking system
295. Library management system
296. Contact management system
297. ATM simulation
298. Tic-Tac-Toe
299. Mini billing system
300. Inventory management system
301. Simple login system
302. To-do list application
303. Password generator
304. Expense tracker
305. File-based student management system
306. Collection-based student management system
307. JDBC-based student management system